

	RUNING COMPLETION	
CATEGORY: REGULAR DOC REF # UC/SWI/DIV/PDHG	ZK460	VERSION: 1.0 ISSUED: 09-07-2026

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Completion Procedure

1. Change **TOP BOP RAMS** to 3 ½" – 5 ½" single variable RAM and test it to 5,000 psi for 30 min.
2. R/U power tong, jam unit, single slips, and elevator.
3. M/U circulation head with low torque valves and keep the same available on rig floor. In addition, injection head and injection manifold should be available on rig floor. **DO NOT CIRCULATE THRU THE TUBING, AS THERE IS GLASS BARRIER DISC IN THE COMPLETION.**
4. Packer assembly shall be run in hole with **INTERWELL pressure multi cycle to open** glass disk at bottom, **autofill of tubing shall NOT be possible**. Ensure to have provision at rig floor to fill the tubing with clean filtered inhibited brine while running in hole.
 - Ensure to use safety clamp with slips ± 50 joints (or enough weight ± 10,000lbs)
 - Pass carefully through the BOP. Completion and packer engineers to be on rig floor to control running speed and observe for any drag
 - Completion engineer(s) to be on rig floor during operation.
5. Conduct pre-job safety meeting prior to running completion.

Make up torque for **4-1/2" 12.6 PPF L80 TSH BLUE**

- **Minimum : 4820 ft-lb**
- **Optimum : 5360 ft-lb**
- **Maximum : 5900 ft-lb**

6. P/U and M/U **sub-assembly #1 SEAL ASSEMBLY)**

SUB-ASSEMBLY # 1 Seal Assembly case		
QTY	THREAD	DESCRIPTION
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5 FT LG PUP JOINT (TOP)
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX	WFD SEAL ASSEMBLY (5.25" SEAL OD)

7. M/U **70 x tubing** joint (**approx. 2,135 ft**) of **4-1/2" 12.6 PPF TSH BLUE** L-80 tubing on top of **sub-assembly # 1**.
8. P/U and M/U **sub-assembly # 2 (BARRIER DISC)**

SUB-ASSEMBLY # 2		
QTY	THREAD	DESCRIPTION
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	7 FT LG PUP JOINT
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5.91" OD INTERWELL GLASS DISC (MULTI CYCLE TO OPEN)
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5 FT LG PUP JOINT

Note: Use safety clamp with slips until gaining enough weight. Approx. 10,000 lbs.

9. M/U **3 x tubing** joints (**approx. 91 ft**) of **4-1/2" 12.6 PPF TSH BLUE** L-80 tubing on top of **sub-assembly # 2**. Fill the tubing.

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Note: Make sure minimum 01 x full joint below packer for setting RBP as contingency. It's preferable to have 02 x full joints unless there is a space out issue.

10. P/U and M/U **sub-assembly # 3 (PACKER). Packer setting depth +/- 14,530 ft.**

SUB-ASSEMBLY # 3		
QTY	THREAD	DESCRIPTION
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	7 FT LG PUP JOINT
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5.91" OD WEATHERFORD OPTIPKR CUT TO RELEASE PACKER
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5 FT LG PUP JOINT

Note:

- Packer engineer, CE & ADNOC DS to check the OD of packer and confirm it is **5.910"**.
- Packer engineer to visually inspect the packer and ensure shear screw numbers in the packer (**2,000 psi** start to set pressure).
- Packer engineer to ensure the location of pip tag is clearly noted and mentioned in tally and schematic from top of the packer and bottom of the packer.
- Take pre-caution while passing through BOP and riser to ensure packer doesn't hang off.
- Control running speed while passing through BOP, and do not exceed 5k lbs. drag.

11. M/U **21** x tubing joints (**approx. 640 ft**) of **4-1/2" 12.6 PPF TSH BLUE** L-80 tubing on top of **sub-assembly # 3**

Note: Permanent Downhole Gauge (PDHG) to be minimum 200 ft above production packer as per WIMS considering future rig-less P&A. **PDHG would not run pass through 7" TOL (13,974 ft), plan PDHG setting depth at +/-13,870 ft**

12. P/U and M/U **sub-assembly # 4 (PDHG MANDREL)**

SUB-ASSEMBLY # 4		
QTY	THREAD	DESCRIPTION
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	7 FT LG PUP JOINT
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	WEATHERFORD GAUGE CARRIER
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5 FT LG PUP JOINT

Note: Follow **WFD PDHG instruction for details of PDHG installation and function test**

13. RIH with 1 x qty of **4-1/2 12.6 PPF TSH BLUE** L-80 tubing above gauge mandrel. Set string until gauge mandrel is at working height to allow installation of gauge.

Note: Install hole cover to prevent any dropped objects down hole.

14. Install PDHG into mandrel as per vendor procedure

- Record baseline resistance of cable before installing PDHG to get reference number

15. Pressure test the gauge connection to 500 psi for 5 mins and **7,500 psi** for 15 mins.

- This pressure test is to ensure no communication between TBG and A annulus. Please ensure test pressure is more than hydrostatic pressure at planned PDHG depth + annulus pressure test
- **Please check with PDHG vendor if we can perform pressure test with this value.**

16. Record gauge reading below to ensure PDHG signal is good

Gauge S/N:	
Pressure Gauge :	psi
Temperature Gauge :	Deg C
Current Gauge :	mA
Voltage Gauge :	V

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17. Install cross coupling clamp on 4-1/2" tubing joints

Note: PDHG engineer and CE to check the OD of each clamp prior to RIH to ensure it's smaller than casing ID

18. M/U **154** x tubing joints (**approx. 4,697 ft**) while installing clamps on 4-1/2" tubing as per the completion tally. Continue filling the tubing every 10 x tubing joints (As tubing is not auto-fill).

- Verify gauge communication continuously and report **every 6 hours** to ADNOC DS.

19. P/U & M/U **sub-assembly # 5.**

SUB-ASSEMBLY # 5		
QTY	THREAD	DESCRIPTION
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	7 FT LONG PUP JOINT
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	3.813" WXN NO-GO NIPPLE @ 50°
1	4-1/2" x 12.6 PPF L80 TSH BLUE BOX x PIN	5 FT LONG PUP JOINT

Note: Make sure inclination is less than 55 degrees for slick line accessibility

20. Continue to RIH with **268 jts** x [4-1/2 12.6 PPF TSH BLUE L-80](#) tubing (approx. **8,174 ft**). Continue filling the tubing every 10 x tubing joints.

Note:

- Completion and packer engineers to be on rig floor to control running speed and observe for any drag.
- Maximum speed is 5 ft. /sec, thus reducing potential damage to the packer.

21. P/U and M/U **sub-assembly # 6 (DHIV Nipple)**

SUB-ASSEMBLY # 6		
QTY	THREAD	DESCRIPTION
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	7' PUP JOINT
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	FLOW COUPLING
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	4-1/2 LANDING NIPPLE WITH 3.813" WX PROFILE
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	FLOW COUPLING
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	5' PUP JOINT

Note:

Confirm no DHIV inside the nipple

22. Continue running **~30-32 Joints 4-1/2" x 12.6 PPF L80 TSH BLUE tubing (DHIV at 1000 ft depth)** while installing cross coupling clamp as per tally. Continue filling the tubing every joint.

- All TBG above DHIV to be drifted by special **3.88"** drift

23. Continue RIH with [4-1/2" 12.6 PPF L80 TSH BLUE](#) tubing joints until **5.25"** OD Seal Stem 30 ft. above [4-1/2"](#) TOL.

24. Record P/U weight and S/O weight, stretch length.

25. Continue to RIH very slowly at a controlled speed and zero the vernier scale. Then slowly continue to RIH cautiously monitoring the vernier scale for any drag indications. Gently tag the top of 4-1/2" liner with the **5.25"** OD Seal Stem of the **sub-assembly # 1**; with a max. of **10 klbs.** slack off weight.

Note:

- Do not tag on the [4-1/2"](#) TOL with more than 10 klbs. The purpose of the tag is to ensure that correct space-out is used and the WLEG is inside the TOL.

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26. P/U to last P/U weight plus 2 ft more and mark the tubing above the rotary table for final space out calculation. Completion engineer to calculate space-out and ensure proper pup joints lengths are utilized

Note:

- Space-out calculation is mandatory to be discussed and agreed with DS and completion engineer in town.
 - Ensure sufficient quantity and length of pup joints are available for the space-out.
27. POOH completion string and L/D excess tubing joints as per completion engineer's space out calculation.
28. Add necessary pup joints below last full tubing joint below tubing hanger while installing cross coupling clamp.
29. P/U and M/U **sub-assembly # 7 (TUBING HANGER)**

SUB-ASSEMBLY # 8		
QTY	THREAD	DESCRIPTION
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	5 FT PUP JOINT
1	4-1/2" L-80, 12.6 PPF TSH BLUE BOX 4-1/2" L-80, 12.6 PPF JFE LION PIN	XOVER PUP JOINT
1	4-1/2 12.6 PPF JFE LION BOX TOP 4.875-4TPI ACME-2G-LH PIN BTM	LANDING SUB
1	4.875-4TPI ACME-2G LH BOX TOP 4-1/2" VAM TOP BOX BOTTOM	CACTUS (BAKER) HANGER ASSY
1	4-1/2, 12.6# VAM TOP PIN X 4-1/2 TSH BLUE PIN	XOVER PUP JOINT
1	4-1/2" L80, 12.6 PPF TSH BLUE BOX x PIN	7 FT PUP JOINT

Note:

- Follow wellhead company program for the details of PDHG cable termination
 - **Drift the landing joint with 3.88" special drift to ensure free passage of 3.81" DHIV.** Ensure landing joint is **free from any scale / cement.**
 - Lubricate the seals of tubing hanger on the outside diameter.
 - Ensure control line fittings for tubing hanger are available with wellhead engineer.
 - Verify the number of control line ports to be run and plug the unused ports.
 - Ensure the lockdown screws on the THS already fully retracted. Measure length of lockdown screws.
30. Take final gauge readings and record results for reference.
31. Cut and strip 20-30 ft of TEC line by removing encapsulation starting from at least 6 ft below the tubing hanger.
- Note:**
- Extra care must be taken not to damage, bend or kink bare tube.
 - Extra care to be taken to not drop any hand tools inside the well (hole cover to be used during stripping / cutting the TEC line).
32. Straightening tool to be used for straightening the TEC line if required
33. Feed 1/4" NPT x Swagelok tube fittings and position below the tubing hanger. Feed the TEC line through 1/4" NPT feed thru ports on the tubing hanger.
34. Feed 1/4" NPT x Swagelok tube fittings over lines from above the tubing hanger.
35. M/U 1/4" NPT fitting above and below tubing hanger. Pull TEC line slack and tighten the Swagelok ferrules to the required torque value.

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36. Attach a Test Pump to the 1/8" LP PORT on the Tubing Hanger OD to test top and bottom tubing sealing of each control lines. Test to maximum 5,000 psi or 80% of control line collapse pressure, whichever is lower. Hold the test pressure for fifteen (15) minutes or as desired by the drilling supervisor. Pressure test also needs to be performed to the un-used control lines port (if any) to ensure there is no leaking on the pipe plugs.
37. Perform 3 wraps of TEC cable around TH neck counterclockwise, bend up to carry till rig floor. Make sure wraps are properly secured.
38. Connect back PDHG cable to original reel and check readings, compare to the readings before splicing & Feed through Hanger.

3.6. Landing Single Tubing Hanger

1. Rig up surface lines to top of **4-1/2 TSH BLUE, 12.6 PPF** landing joints. Allow enough slack in chick-sans for all pipe movements.
2. P/T surface lines to 4,500 psi for 15 mins.
3. Perform temporary splice on the TEC cable to reconnect back to Data Acquisition Unit to monitor and log gauge data while landing TH and setting the packer. Any abnormalities during operation must be reported to Drilling Site Leader.
4. Take a TDR trace prior to landing the tubing hanger to check cable integrity
5. Drain the riser through THS side outlet valve, keep the valve open to prevent pressure lock while landing the tubing hanger.
6. Ensure lockdown screws on tubing head spool already fully backed out
7. Run tubing hanger with landing sub through the rotary table into the THS.
8. Lower the completion string slowly watching TEC Line to confirm proper space-out.
9. Confirm the electronic gauge is working, while lowering the tubing hanger.

Note:

- PDHG Engineer and Completion Engineer to be present on rig floor while lowering the tubing hanger.
10. Measure and mark the drill-pipe joints while descending. Land completion string inside THS as per WH company procedure. Confirm final space-out based on no-go tag depth when landing tubing hanger. This should be supervised and confirmed under WH engineer's supervision.
 11. After landing the tubing hanger, confirm that gauge is functioning as per the specs and is recording the data.
 12. Tighten the LDS in crisscross pattern to energize tubing hanger elastomeric packing. Confirm correct final engagement standoff of LDS to be max **2.45"**. Recommended torque **650 ft-lbs.** (above rotational torque).
 13. Perform pressure test on the Tubing Hanger OD seal by applying pressure through HGR SEAL TEST PORT on the Multibowl to maximum 5,000 psi. Hold for 15 minutes.

Note : This is to ensure TBG HGR integrity before setting packer

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3.7. SET & TEST COMPLETION

- Slowly pressure up the tubing to **500** psi against the **INTERWELL** Glass Disc and hold for **5** min.

Note : This is to ensure TBG integrity before setting packer

- Increase pressure to **4,100** psi continuously and hold for **30 min**. Initial packer setting shear pressure may occur at **1,500** psi.

DESCRIPTION	FUNCTION	THEORITICAL VALUE
PACKER SETTING PRESSURE	MIN SETTING PRESSURE	2,000 PSI
	FULL SETTING PRESSURE	4,000 PSI
	HOLD POINT (TBG INTEGRITY TEST)	4,100 PSI

Note:

- In case of issue (leak during P test), Check surface line for any leaks.
 - Consider using nipple-less plug and set in tubing between packer and glass disc.
- R/UP cementing lines to THS SOV, and test A-annulus below hanger to **500** psi and hold for **5** minutes, then increase pressure to **3,500** psi in stages and hold for **30** min. Keep the tubing open and filled up to the top with completion brine and observe for pressure drop and any returns during annulus testing. Record the test and send to town.
 - Reverse Ballooning may cause some returns from the tubing string for few minutes. If returns do not stop after stabilizing the pressure, call DROTL.
 - Monitor tubing strings for any returns.
 - Close THS inner SOV and bleed pressure in line to **500** psi (inflow test of the side outlet valve). Monitor for **15** min.
 - Increase pressure back to **3,500** psi then bleed off annulus pressure slowly to zero, then open THS outlet valves.

Note: SOV inflow test can be waived if it's previously done during THS installation

4. N/D BOP & N/U X-MAS

4.1. N/D BOP

- Make sure the following SL pre-check is completed
 - Test surface lines and slick-line lubricator @ 4,500 psi for 15 min as per standards prior to first slick-line operation
 - Perform torture cable test for 0.125" S/L. Provides 17 rounds, if breaks at or after 17 rounds then the slick-line is OK. Proceed with normal slick-line operations. If slick-line breaks before 17 rounds. Follow step b)
 - Proceed to cut 100' of cable and perform torture cable test once again. Cable must break at or after 17 rounds, otherwise cut 100' and repeat test again, until cable breaks at 17 turns or after.
 - Every 45 mins of jarring rope socket should be re-cut and fresh knot should be created.
 - All downhole tools length and dimensions shall be measured, recorded and reported to CE and DROTL in town.
 - Fish neck profile and dimensions shall be documented and reported for all BHA component prior to RIH.
 - All BHA proposals to be shared with ADNOC Offshore Completion Engineer in town

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2. Run and set **3.81" PX** blanking plug on **3.81" WX** Nipple, pressure test plug to **2500** psi
3. Well has two downhole barrier as per below:
 - **Primary Barrier:** Barrier disc (Pressure tested)
 - **Secondary Barrier:** 3.81" blanking plug (Pressure tested)
4. **Back off L/D landing joints from string.**

Note: The landing joint connection is 4.875-4TPI ACME-2G (Left Hand Thread)
5. Once landing joint is retrieved, record the surface voltage and current, pressures and temperatures and power down the PDHG.
6. Install 4" H2 valve at the tubing hanger.
7. Lift BOP and carefully disconnect TEC line and make sure not to damage the wraps of cable while pulling.
8. Check the s-seal on the tubing hanger extended neck for any damage and ensure the neck stick-up height is as shown on the drawing.
9. Maintain control line 3 rounds and TEC line 3 rounds around the extended neck of the tubing hanger. Make sure enough length is provided for termination at the TH control line exit ports.
10. For TEC line, feed the cable through the THS and install the wellhead outlet and perform Pressure test to **5,000** psi for 15 min as per PDHG vendor procedure.

4.2. XMAS TREE INSTALLATION

Preparation & installation

- All Xmas tree valves to be pressure tested by offline except inflow test of lower master valve
 - Verify and record the part number and serial number of the Tree (XT) and its accessories.
 - Ensure that XT sub-components are complete as per attached procedure and drawing.
 - Drift the XT bore with 4" drift.
 - Inspect the XT bottom seal prep and seal sleeve S-seals for visible damage. Lightly oil the sealing areas.
 - Inspect and clean the **RX-54** ring grooves on the THS and XT.
1. Place ring gasket **RX-54** on the THS upper ring groove.
 2. Attach bleeder plug stinger tool to bleeder plugs on XT test ports (cavity and seal test ports). Keep the ports open to avoid any pressure lock during installation.
 3. Fill cavity of THS with light oil and lubricate Seal Sleeve OD.
 4. Install the seal sleeve assembly onto the THS.
 5. Pick up XT vertically with a certified sling. Orientate the XT according to ADNOC offshore requirement.
 6. Orientation & Height of XT should be confirmed by ADNOC Offshore CED team.

Note:

- Follow Wellhead company instructions to install XT.
- Never use hemp or nylon rope or chains to suspend or move XT. Failure to use wire rope could result in dropping the tree and causing severe injury to personnel.

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- While landing XT onto THA, keep it as centralized as possible to prevent contact and damage to the Seal Sleeve.
 - 7. Remove bleeder tools from XT cavity test ports.
 - 8. Hook up test lines to pressure test cavity to **5,000** psi for 15 mins through the cavity test port of XT. Upon successful pressure test, bleed off completely. Leave the XT cavity test port open.
 - 9. Remove bleeder tools from XT seal test ports.
 - 10. Pressure test S seals to **5,000** psi for 15 minutes through one of the two XT S Seal test ports.
 - 11. Repeat steps for the second S Seal test port located 180 degree away.
 - 12. Line up cement line from cement unit to XT kill valve. XT shell test to be recorded on chart at cement unit.
 - 13. **Retrieve H2 valve from the tubing hanger.**
- Note: To be reported clearly in DDR & wellhead company report**
- 14. R/U SL and run **4" SB** pulling BHA to recover prong and run **4" GR** pulling BHA to recover **3.81" PX** blanking plug
 - 15. Pressure up the completion string to **4,000** psi (against the barrier disk device) and hold for **15** mins for **shell test**, close lower master valve (LMV) and bleed off the pressure above it to **3500** psi (**Low inflow test of LMV**) monitor **15** minutes, then bleed off pressure above LMV to **500** psi (**High inflow test of LMV**) and monitor **15** minutes for any increment in the pressure.
 - 16. Pressure up above LMV to **4,000** psi to equalize pressure across the LMV. Open LMV and bleed off string pressure to 0 psi.

5. OPEN BARIRER DISC AND SECURE WELL

5.1. Open Barrier Disc by Pressure Cycle

1. Confirm remaining cycles
2. Pressure up TBG to **3,500** psi and fold for **2** minutes
 - No specific requirement for pump rate to pressure up as long as it is steady rate.
3. Beed down TBG pressure to 0 and fold for **2** minutes
4. Repeat step #3 & #4 for **11** more cycles
5. At the end of the last cycle while bleeding off to 0, barrier disc should be opened

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5.2. Bull-heading SW with Injectivity Test

1. Bullhead sea water at maximum 0.7 psi/ft for (TBG volume + LC volume = 560 bbls)
 - SPP is limited to 0.7 psi.ft
 - Record SITHP
2. Report pump rate with surface pressure in DDR

Note:

- Open annulus A during injectivity test and check for returns to confirm the permanent packer integrity

5.3. SECURE WELL

1. Record & Report Well head Shut in pressure.
2. **R/U SL unit and Run 3.81" WRDHIV and set on 3.81" WX profile as per SL company procedure and perform inflow test against WHSIP**
3. Secure well under DS supervision by closing all X-Mass tree valves except LMV and prepare for rig skid to next slot

*****END*****

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6. APPENDIX

6.1. CONTINGENCY: Mill barrier disc if unable to remote cycle open.

Notes:

- Ensure to have acid carboy available in the event of any stuck incident.
- Ensure to have strainer on surface line in order to prevent any debris inside CT.
- Flush Pumping unit & surface line prior to connecting with CT

CT OPERATION FOR MILLING GLASS DISC		
CT OPS NO: 1	COMP TUBING	MILLING GLASS DISC
• Install shear seal on top of the X-Mass tree, install CT riser, and BOP. • Test CT against closed swab valve to 5,000 psi. M/UP CT injector, flush string and test 1.5" coil tubing reel to 4,500 psi.		
• M/U CT connector and perform pull test to 25 k • M/U 2.5" Weatherford Milling BHA with 2-1/8" motor on CT.		
• RIH as per CT company procedure, fill reel volume with 12.1 ppg brine while breaking circulation every 1,000 ft. and recording weight every 2,000 ft.		
➤ Ensure not to exceed motor BHA activation pressure during break circulation		
• RIH CT as per Through TBG company procedure and fill reel with 12.1 ppg brine while breaking circulation every 1,000 ft & recording weight every 2,000 ft.		
• RIH CT @ 50 ft. /min. Reduce running speed to 15 ft./min 100 ft. above and below all restrictions and changes in ID as per completion schematic		
• Take P/UP and SO weight 50 ft. prior to reach glass disc.		
• Continue RIH with mill assembly and mill the disk according to Through Tubing company instructions.		
• Once glass disc has been milled, pass through 3 times (15' below and above) with milling BHA to confirm complete removal. Circulate out 1.5 tbq. Volume + CT volume with 8.6 ppg seawater to circulate out any debris.		
• POOH CT to surface & R/D CT.		
➤ After CT BHA at surface, bullhead diesel for remaining lower completion volume		

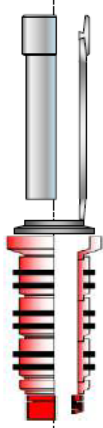
6.2. CONTINGENCY: To Set RBP below production packer (for hydraulic set production packer)

4-1/2" RBP will be set either by Coiled Tubing or E-line Tractor. Separate procedure will be issued by wireline and RBP services company.

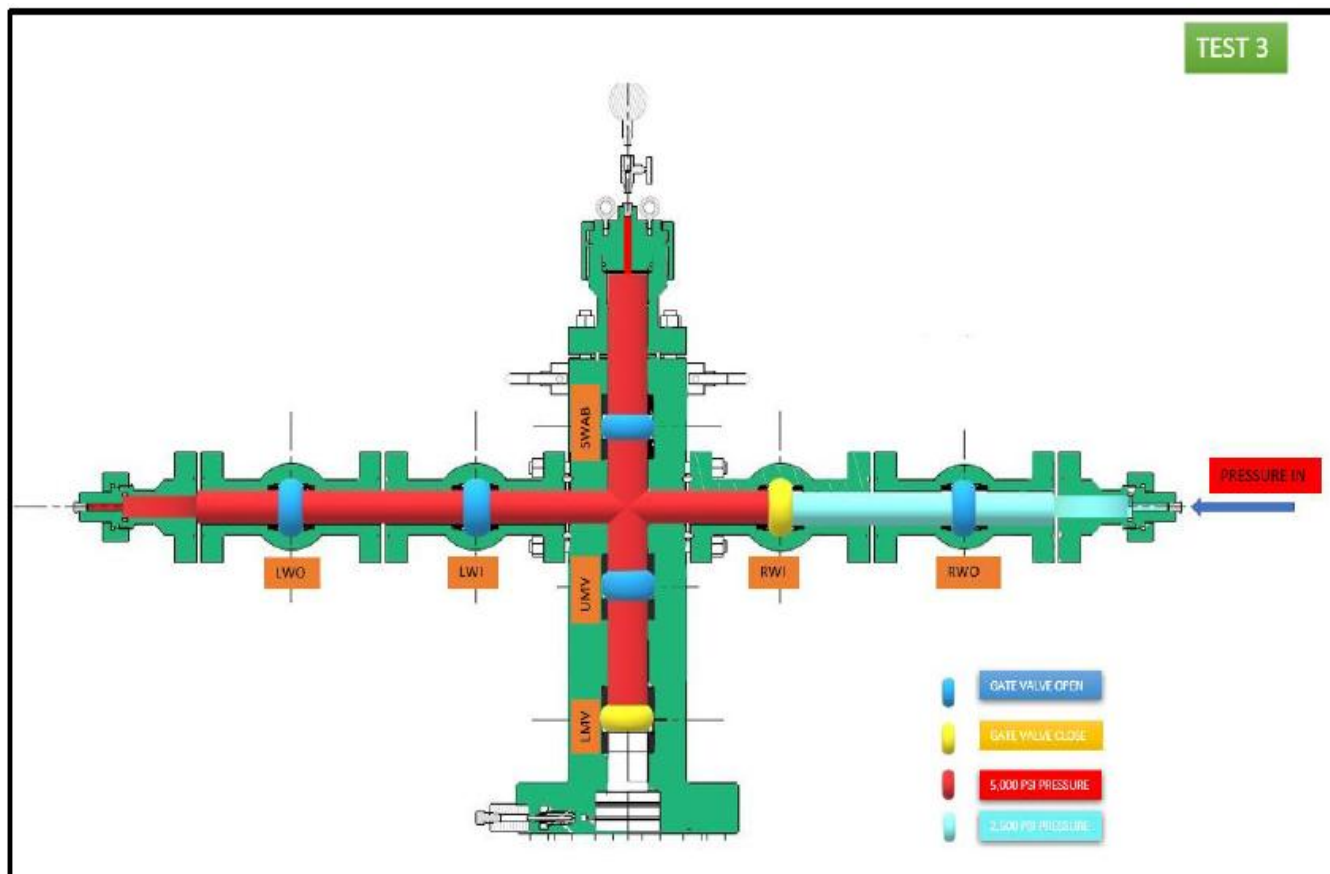
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7. ATTACHMENTS

7.1. Completion Sub Assembly Report

		ADNOC OFFSHORE									
Completion Sub-Assembly Drawing											
Well No:		ZK-460				Rig:		AL SILA			
Sub Assembly:		ASSEMBLY No. 1				Date:		7/1/2026			
Completion Engineer:		HAKMIN ADOW									
Sub Assembly Supervisor:		NIBRAS BITAR									
ASSEMBLY DRAWING	ITEM	DESCRIPTION	LENGTH (FT)	O.D.	I.D.	TORQUE VALUE (ft/lbs)	ACTUAL TORQUE (ft/lbs)	CHART MAKE-UP#	DRIFT		
	1	5' PUP JOINT 4-1/2" L80 12.6 PPF TSH BLUE BOX PIN SAP MN: 130010000925	4.82	4.960 4.500	3.958				3.883		
						5360	5361	19			
	2	SUB, B-B5.00 18.00#VTHT BOX X 4.5 12.6 #TSH BLUE L80 BOX PN:3160469 SN:12380013-1	2.34	5.631	3.863				3.833		
						7960	7965	18			
	3	WEATHERFORD SEAL STEM TBSS16-15 5.0 18.00# VTHT PIN(5.250 PBR ID) L80, SUB, B-B 5.00 18.00# VTHT X 4.50 12.60# TSH BLUE L80 (WS302) SAP MN: 130010043543 PN:3161342 SN:12380013-1 6 seal	18.58	5.776	4.181				3.883		
		TOTAL LENGTH OF ASSEMBLY	25.74								
TORQUE MACHINE S/N:		CE001206-TM RT+22 120				TORQUE VALUES					
CHART RECORDER S/N:						Size	Weight	Thread	Min	Optimum	Max
PRESSURE TEST INSTRUCTIONS:						3-1/2"	9.2	VAM TOP	2,610	2,900	3,190
ASSEMBLED BY:		WEATHERFORD				4-1/2"	12.6	VAM TOP	4,000	4,440	3,817WPX N
TESTED BY:						4-1/2"	12.6	TSH BLUE	4,820	5,360	5,900
DATE:		23-06-2026									
STEPS TO BE COMPLETED AT SUB ASSEMBLY:											
1	ASSEMBLY MADE-UP AT WEATHERFORD WORKSHOP COMPLETIONS UNDER ADNOC OFFSHORE SUPERVISION								YES	NO	
4	MAKE-UP THE ASSEMBLY CONNECTIONS TO THE OPTIMUM TORQUE								☐	☐	
5	ASSEMBLY DRIFTED AFTER THE MAKE UP WITH 3.883" API DRIFT(SN:87579069-2)								☐	☐	
6									☐	☐	
7									☐	☐	
8									☐	☐	
9									☐	☐	
10									☐	☐	

	UPPER COMPLETION INSTRUCTIONS (SWI)	
CATEGORY: REGULAR DOC REF # UC/SWI/DIV/PDHG	ZK460	VERSION: 1.0 ISSUED: 09-07-2026



Testing Instruction:

1. Close the LMV
2. Open UMV, LWO, LWI, SWAB, RWI and RWO.
3. Pressurize the X-mas tree bore to **Test Pressure** for 15 min.
4. Close the RWI and Drop the Pressure line to 500 PSI.
5. Monitor the pressure line for any build up pressure than 500 PSI and hold the pressure for 15 mins.
6. After successful test, Pressurize the line to Test Pressure and Open the RWI valve
7. Release the pressure to Zero PSI

Note: this pressure test will ensure the sealing of gate and seat for the RWI

		UPPER COMPLETION INSTRUCTIONS (SWI)	
CATEGORY: REGULAR		ZK460	VERSION: 1.0
DOC REF # UC/SWI/DIV/PDHG			ISSUED: 09-07-2026

7.18. Coil Tubing Accessibility



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NOWMO Petroleum Services L.L.C.

Coiled Tubing Simulation Analysis

- Summary Report -

Client	ADNOC Offshore	Field	Lower Zakum	Well Number	ZK-104
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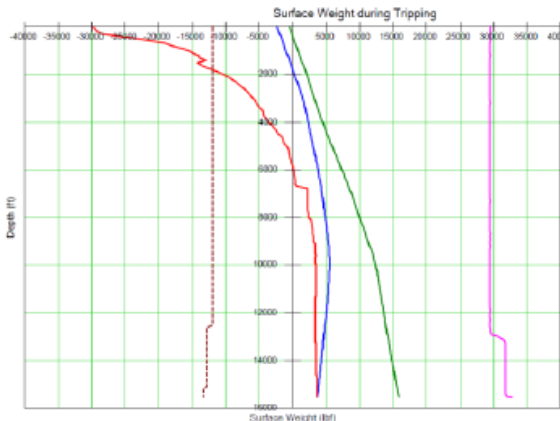
Well Total Depth	25500.0 ft				
Csg / Liner Size	4.5 in	Tubing Size	4.5 in	Open Hole Size	6.0 in
Csg / Liner Depth	25500.0 ft	Tubing Depth	16813.0 ft	Op. Hole Length	0.0 ft

Coiled Tubing Accessibility

CT Size (in)	Depth Reached (ft)	Off Bottom (ft)	Fluid Weight (ppg)		BHA Description	Friction Coefficient		% Well Accessed
			In Well	In CT		Tubing	C.H/OH	
1.5	16,794	8706	12 ppg Brine	12 ppg Brine	Standard BHA	0.3	0.3	66%
1.5	14,198	11302	12 ppg Brine	12 ppg Brine	Standard BHA	0.4	0.4	56%
1.5	18,278	7222	12 ppg Brine	12 ppg Brine	Agitator BHA	0.3	0.3	72%
1.5	15,558	9942	12 ppg Brine	12 ppg Brine	Agitator BHA	0.4	0.4	61%
1.75"	18,100	7400	12 ppg Brine	12 ppg Brine	Standard BHA	0.3	0.3	71%
1.75"	14,909	10591	12 ppg Brine	12 ppg Brine	Standard BHA	0.4	0.4	58%
1.75"	19,781	5719	12 ppg Brine	12 ppg Brine	Agitator BHA	0.3	0.3	78%
1.75"	16,333	9167	12 ppg Brine	12 ppg Brine	Agitator BHA	0.4	0.4	64%
2"	18,650	6850	12 ppg Brine	12 ppg Brine	Standard BHA	0.3	0.3	73%
2"	15,711	9789	12 ppg Brine	12 ppg Brine	Standard BHA	0.4	0.4	62%
2"	19,869	5631	12 ppg Brine	12 ppg Brine	Agitator BHA	0.3	0.3	78%
2"	16,813	8687	12 ppg Brine	12 ppg Brine	Agitator BHA	0.4	0.4	66%

Coiled Tubing Operating Limits

Well Intervention Simulation for Coiled Tubing



Note : The lockup limit and the 50% Yield strength are not to be exceeded at any time.

Note : Detailed CERBERUS software simulation report to be attached to this summary sheet.

Note : Simulations performed are based on hypothetical friction factors. Results may vary based on actual well conditions during operation.

*****END*****